

|          |                                                                         |           |
|----------|-------------------------------------------------------------------------|-----------|
|          |                                                                         |           |
| 5(a)(i)  | (two) waves meet/overlap (at a point)                                   | <b>B1</b> |
|          | (resultant) displacement is sum of the displacement of each wave        | <b>B1</b> |
| 5(a)(ii) | constant phase difference (between the waves)                           | <b>B1</b> |
| 5(b)     | $I \propto A^2$<br>$3I / I = (A + 1.5)^2 / 1.5^2$                       | <b>C1</b> |
|          | $A = 1.1 \text{ cm}$                                                    | <b>A1</b> |
| 5(c)(i)  | $\lambda = ax / D$                                                      | <b>C1</b> |
|          | e.g. $a = 680 \times 10^{-9} \times 2.0 / 4.0 \times 10^{-3}$           | <b>C1</b> |
|          | $a = 3.4 \times 10^{-4} \text{ m}$                                      | <b>A1</b> |
| 5(c)(ii) | straight line from positive value on x-axis and always below 'old' line | <b>B1</b> |
|          | straight line with a smaller positive gradient than 'old' line          | <b>B1</b> |

| Question | Answer | Marks |
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